

KNOWLEDGE TRANSFER DOCUMENT

BUIDCO Project Monitoring Dashboard

A beginner's guide to what the system is, who uses it, and how every feature works

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1. What is this project? (The 2-minute explanation)

BUIDCO runs many infrastructure construction projects at once — roads, drainage, water supply schemes, buildings, and similar public works — spread across multiple districts, divisions (regional engineering offices), and funding schemes. Historically this kind of information tends to live in scattered Excel sheets, which makes it hard for leadership to answer simple questions quickly: *How many projects are delayed? Which contractors' guarantees are about to expire? Which projects are behind on progress compared to their money already spent?*

This application is a **web-based dashboard that replaces those spreadsheets**. Project Directors (site-level engineers) enter and update project information through a structured form. The system then automatically rolls that information up into dashboards, alerts, and reports so that senior management (the MD — Managing Director — and Admins) can monitor the entire project portfolio at a glance, drill down into any single project, and see a full history of who changed what.

In one sentence: **it is a single source of truth for "what is the status of every project we're running, and is anything about to go wrong?"**

2. Who uses the system, and what can they do?

There are four types of user accounts (**roles**). Every screen and every API request is restricted based on which role is logged in.

Role	Who they are	What they can do
MD	Managing Director — the top of the organisation	Sees and can edit everything , across all divisions. The only role that can open the Audit Trail (full change history) and manage all user accounts, including creating other Admins.
Admin	Head-office staff who administer the system	Full access to project data across all divisions (create/edit/delete). Can create and manage Viewer and PD accounts, but cannot create another Admin or MD.
PD	Project Director — a site/division-level engineer	Only ever sees projects belonging to their own division(s) — this is enforced by the server itself, not just hidden in the menu. What exactly they can edit is controlled by individual permission flags (see below).
Viewer	Anyone who just needs to look at the data	Read-only by default. An Admin/MD can optionally grant a specific Viewer permission to create, update, or delete project records — those four permissions (<code>canViewProjects</code> , <code>canCreateProjects</code> , <code>canUpdateProjects</code> , <code>canDeleteProjects</code>) are set individually per user.

Non-project features — Minutes of Meeting, Change of Scope/EoT requests, Management Actions, Pre-Monsoon checklist — use a simpler rule: only Admin and MD can add or edit; everyone else can only view.

3. How the pieces fit together (architecture, in plain English)

The system is a classic three-layer web application:

Layer	What it is	Technology used
Frontend (what you see in the browser)	The dashboard, forms, tables, and charts you interact with.	React + TypeScript, built with Vite; Redux Toolkit for state/data-fetching; Tailwind CSS for styling.
Backend (the "brain")	Receives requests from the browser, checks who you are and what you're allowed to do, talks to the database, and calculates all dashboard numbers.	Node.js + Express + TypeScript, using Drizzle as the ORM (the translation layer between TypeScript code and SQL).
Database (where the data lives)	Every project, user, meeting, photo, and audit record is permanently stored here.	PostgreSQL.

When you click something in the browser, the frontend sends a request to the backend (e.g. "give me all projects in Patna division"). The backend checks your login token, figures out your role and permissions, asks the database for the data, shapes it into a JSON response, and sends it back for the frontend to display. Login sessions use a short-lived **access token** (kept only in browser memory, expires in 15 minutes) plus a longer-lived **refresh token** stored in a secure cookie that silently renews the access token — so you don't get logged out just because 15 minutes passed, but a stolen access token has a very short useful life.

4. Glossary — terms you'll see everywhere

Term	Meaning
AA	Administrative Approval — the officially sanctioned budget for a project.
PBG	Performance Bank Guarantee — a bank guarantee a contractor deposits as security; it has an expiry date the system tracks and alerts on.
EMD	Earnest Money Deposit — a good-faith deposit paid by a contractor when bidding for/signing a contract.
CoS	Change of Scope — a formal, approved change to what a project covers or its cost.
EoT	Extension of Time — a formal, approved extension of a project's deadline. CoS and EoT are usually requested and approved together.
O&M	Operation & Maintenance — the period after construction finishes where a contractor is still contractually responsible for maintaining the asset.
MPR	Monthly Progress Report — a monthly snapshot of funds received, expenditure, and manpower on a project.
RA Bill	Running Account Bill — an interim payment bill a contractor submits as work progresses (as opposed to one lump payment at the end).

Milestone	A defined chunk of a project's work (e.g. "Foundation") given a weight (e.g. 20% of the whole project); progress against milestones is what the system uses to compute an accurate overall physical-progress percentage.
Division / Region	The organisation's geographic structure: several Divisions (engineering offices) sit inside a Region. Every project belongs to one Division, and every PD user is assigned to one or more Divisions.
Scheme	A government funding programme. One project can be funded under more than one scheme.
Sector	The category of infrastructure a project is (e.g. Water Supply, Roads, Buildings).

5. The data model — what information the system tracks

Everything in the system revolves around one central record: the **Project**. Think of the database as a set of connected spreadsheets (tables); the Project sheet is the main one, and every other sheet either describes the Project further or classifies it.

The central record: Project

One row = one construction/infrastructure work. It stores: name, sector, city/district, which division it belongs to, the contractor's name, the assigned PD, current stage and status, all the money fields (sanctioned cost, agreement value, advances, retention, PBG, EMD, payments made), physical vs. financial progress percentages, planned/revised/ expected completion dates, and O&M info for after construction ends.

Reference / classification tables (dropdown lists)

- **Region** → contains several **Divisions**
- **District** — where the project physically sits
- **Sector** — the type of infrastructure
- **Scheme** — funding programme (a project can have more than one, via a linking table)

Records that belong to one specific project

Table	Real-world meaning
Project Milestone + Milestone Progress	The weighted work-breakdown for a project, and the monthly % progress recorded against each milestone.
CoS/EoT Item	A formally approved scope-change and/or deadline-extension event for the project.
Management Action Item	A to-do/follow-up item raised against the project, with a status (Open/Closed) and deadline.
Geo Photo	A geotagged site photo (uploaded file or link) with a caption and date — visual proof of progress.

Records that stand on their own (not tied to one project)

Table	Real-world meaning
Pre-Monsoon Item	A corporation-wide checklist task to prepare for monsoon season (e.g. clearing drains), with priority and deadline.
Minutes of Meeting (MoM)	A record of a meeting — date, venue, chairperson, attendees, agenda, decisions — optionally linked to one project.

MoM Action Point	An individual follow-up task that came out of a specific meeting, with an owner, due date, and status.
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Accounts & accountability tables

`app_user` (login accounts and their permission flags), `user_division` (which division(s) a PD is assigned to), `refresh_token` (active login sessions), and `audit_log` / `audit_log_change` (the full change history — see Section 10).

6. Logging in — including the special Project Director flow

The login screen is a simple username + password form. What happens next depends on the role:

- **MD, Admin, Viewer** — credentials are checked, and the user is signed in immediately, landing on the Overview dashboard.
- **PD — a two-step flow** — because one PD can be responsible for more than one division, the system can't yet decide which division's data to show. So:
 1. Step 1: the PD enters username + password. The server verifies them but does *not* log the user in yet — instead it replies with the list of divisions that PD is assigned to.
 2. Step 2: the login screen swaps to a dropdown of those divisions. The PD picks one and submits again. Only now is a session actually created — and it's locked to that division for the rest of the session. Every project list, KPI number, and dashboard chart the PD sees for the rest of that session is automatically filtered to their chosen division, enforced on the server (a PD cannot see another division's data even by tampering with the browser).

If the logged-in user is an MD, a one-time "MD Scheme Summary" briefing pop-up appears right after login, giving a quick portfolio snapshot.

7. Page-by-page walkthrough of the app

Once logged in, every page is reached from the left-hand sidebar and shares the same top navigation bar.

Overview (home page, everyone)

The portfolio-wide dashboard. At a glance you see:

- **PBG Expiry Banner** — a red alert bar (only shown when relevant) listing projects whose bank guarantee expires within 30 days.
- **KPI Grid** — total project count broken down by status (Completed, In Progress, Delayed, On Hold, Not Started), plus totals for sanctioned/agreement/ financial value and average progress percentages.
- **Stage Buckets** — how many projects sit at each lifecycle stage (Concept → Pre-Tender → Tender → Construction → O&M).
- **Schedule vs Actual** — compares planned progress to real progress, flagging whether the portfolio is ahead or behind schedule overall.
- **PBG Alerts** and **O&M Alerts** — lists of individual projects with guarantees or maintenance contracts nearing expiry.
- **Work Type Counts, Sector Snapshot, District Summary, Financial Securities** — smaller summary widgets giving quick counts and totals from different angles.

Projects (everyone with view rights)

The full register of every tracked project as a searchable, filterable, sortable table. Clicking a row opens that project's read-only **Project Detail** page (status and priority badges, progress bars, key metrics), which has an "Edit in Input Sheet" button for users allowed to make changes.

Sectors / Schemes / Districts / Divisions (Districts & Divisions hidden for PD)

Four different ways of slicing the same project data — by infrastructure type, by funding programme, by district, or by engineering division. Each shows summary cards you can click to drill into the underlying list of projects. Divisions also shows a region-level roll-up strip above the division cards.

CoS / EoT (everyone with view rights)

A master list, across every project, of Change-of-Scope and Extension-of-Time events — useful for seeing which projects have had their cost or deadline formally revised, and by how much.

Management Action (everyone with view rights; Admin/MD can add)

A cross-project view of follow-up action items, with tabs for "projects with open actions" vs. "projects with no actions" — a quick way to see what still needs chasing.

Outstanding Gaps (everyone with view rights)

A table of every project that has been explicitly flagged (from the Input Sheet) as having an unresolved issue, filterable by priority — a "problem list" for leadership.

Pre-Monsoon Prep (everyone with view rights; Admin/MD can add)

A checklist of tasks that need doing before monsoon season each year (e.g. drain clearance), independent of any single project, with priority and deadlines.

Minutes of Meeting & O&M (reached via top nav, not the main sidebar)

MoM is a log of meetings (date, venue, chairperson, attendees, agenda, decisions) with their resulting action points (owner, due date, status). **O&M** tracks post-completion maintenance contracts and their expiry status.

Audit Trail (MD only)

A page-through log of every change made anywhere in the system — see Section 10.

User Management (MD and Admin)

Create accounts, assign roles, assign PDs to their division(s), and toggle the four individual project permissions for Viewer/PD accounts.

8. The Input Sheet — the big data-entry form, explained section by section

This is where a project's information actually gets typed in or updated — reached from "Add Project" or the "Edit" button on a project's detail page. It's split into two groups of tabs: **Fixed Inputs** (locked to MD/Admin only — a PD can view but not edit these) and **Variable Inputs** (editable by anyone with update rights, including PDs for their own division's projects). Only the project name is strictly required to save.

Section	Group	What gets entered
Basic Info	Fixed	Project name, sector, city/district, region & division, contractor, assigned PD, scheme(s), work description, contract type, sponsoring department, implementing agency, sanction date, project brief.
Contract & Security	Fixed	Agreement number/date, contract value, mobilization advance issued/recovered, PBG number/amount/expiry/bank, EMD, total payments made, last RA bill, retention money — all in ₹ Crore.
Phase & Dates	Variable	Project stage (Conceptualisation → Design → Tender → Construction → O&M), execution status, planned/revised/expected end dates, and an auto-calculated delay-in-days figure that accounts for any approved EoT.
Progress & Financial	Variable	Physical progress % (actual vs. scheduled), financial progress %, AA amounts, and an optional Monthly Progress Report block (funds received, expenditure, manpower, remarks).
CoS / EoT	Variable	A repeatable list — each entry pairs one scope change (category, amount, % variation) with its time extension (days granted, old vs. new end date). This is the actual mechanism by which a deadline gets formally revised.
Geo-Tagging	Variable	A reference URL for the project plus a gallery of site photos, added by pasting a link.
Action & Remarks	Variable	Marks whether the project has an "Outstanding Gap" (with a description, which then surfaces on the Outstanding Gaps page), sets a priority, and lets users add/close Management Action items.
O&M Details	Variable	Whether O&M applies, start date, period in months, agency, and status — auto-computed as Ongoing / Expiring Soon / Expired from the dates, with a manual override available.
Milestones & Progress	Variable	Define weighted milestones (must sum to 100%), then log monthly % progress against each — this feeds directly into the project's overall completion percentage shown everywhere else in the app.

9. How dashboard numbers are calculated (KPIs)

Every number shown anywhere in the dashboards is computed in a single, shared place in the backend (deliberately — so that MD/Admin portfolio-wide figures and a PD's division-scoped figures are always calculated the same way and never drift apart). Beginners should think of this as the "reporting engine." What it produces:

- **Status breakdown & totals** — counts by status, total sanctioned/ agreement value, average physical & financial progress.

- **Financial utilisation %** — money actually spent divided by money sanctioned; the classic check for "is spending matching actual work done?"
- **Schedule vs. actual progress** — planned progress % vs. real progress %, to flag projects falling behind.
- **Delay tracking** — for each delayed project: total delay days, how many of those days are "covered" by an approved EoT, and how many are unexplained/uncovered — an important red flag for oversight.
- **Financial securities totals** — mobilization advances outstanding, retention money held, PBG/EMD totals, and a count of expired or soon-to-expire PBGs.
- **Alerts** — PBG expiring within 30 days; O&M contracts nearing expiry.
- **Breakdowns for charts** — by scheme, sector, district, division, and region, each with counts by status and average progress.
- **Per-project history** — monthly progress trend and milestone-level history, used on an individual project's detail page.

Every one of these calculations can be run "for everyone" (MD/Admin view) or "for just one division" (PD view) — a PD logged into a specific division only ever sees numbers re-calculated from their own division's projects, enforced by the server.

10. Audit trail — accountability for every change

Every time a project, user, or related record is created, updated, or deleted, the system writes a permanent record of it *in the same database transaction* as the actual change — so if the save fails, nothing gets logged either, and vice versa. Two things are recorded:

- **Who / what / when** — the user who made the change, what action it was (Created / Updated / Deleted), and which project it affected (with the project's name saved as a snapshot, so the audit entry still reads clearly even if that project is later renamed or removed).
- **Field-by-field changes** — for updates, exactly which fields changed and their before/after values (e.g. "Sanctioned Cost: ₹5 Cr → ₹6 Cr"). Only fields that actually changed are recorded, and sensitive/bookkeeping fields (passwords, tokens, timestamps) are never included.

This gives a complete, tamper-evident history of who changed what and when — important for a government project where money and deadline changes need to be traceable. Only the MD role can view this trail, from the Audit Trail page.

11. Behind the scenes — the backend API

For whoever picks up development work, here is the backend's route map — one file per feature area, each pairing an Express route file with a "service" file that contains the actual business logic and database queries.

Route file	Feature it supports
<code>auth.ts</code>	Login (incl. the PD 2-step flow), token refresh, logout, "who am I".
<code>projects.ts</code>	Core project CRUD; also mounts CoS/EoT, management-actions, milestones, and geo-photos as nested routes under a specific project.
<code>cosEot.ts</code>	Change of Scope / Extension of Time requests.
<code>geoPhotos.ts</code>	Geotagged site photos.
<code>managementActions.ts</code>	Action items/to-dos against a project.
<code>milestones.ts</code>	Weighted milestones and monthly progress against them.
<code>mom.ts</code> / <code>momActionPoints.ts</code>	Minutes of Meeting and the action points raised in them.
<code>preMonsoon.ts</code>	Pre-monsoon preparedness checklist.
<code>users.ts</code>	User account management.
<code>lookups.ts</code>	One endpoint returning all dropdown reference data (districts, regions, divisions, sectors, schemes).
<code>kpis.ts</code>	All dashboard/reporting numbers (Section 9); automatically division-scoped for PDs.
<code>audit.ts</code>	MD-only endpoint to page through the audit log.
<code>uploads.ts</code>	File uploads, proxied through the UploadThing service.
<code>health.ts</code>	A simple liveness check for monitoring — no business meaning.

Access control is layered: a JWT access token identifies who you are; middleware then checks your role (MD/Admin/PD/Viewer), your individual permission flags for project actions, and — for PDs — whether the project you're touching belongs to your assigned division (if not, the server returns a 404, not a 403, so a PD can't even tell whether another division's project exists).

12. Running the project locally (for developers)

1. PostgreSQL must be running, with a database created for the project and connection details in `backend/.env.local` (see `backend/.env.example` for the full list of variables — database URL, JWT secrets, allowed CORS origins, etc.).

2. From `backend/` : `npm install` , then `npm run db:migrate` to create all tables, then `npm run dev` to start the API on port 4000.
3. From `frontend/` : `npm install` , then `npm run dev` to start the web app on port 5173 (its dev server automatically forwards `/api/...` calls to the backend on port 4000).
4. A first login account can be created with `npm run db:seed-admin -- --username <name> --password <pw> --role MD` from the `backend/` folder.